LAB REPORT

Lab Title	Data Files	Lab No.	09
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Lab description/objectives:

1. Demonstrate understanding of the data files in C programming.
2. Implement functionality of File Data Filtering and Average Calculation as defined in the lab tasks/requirements.
3. Analyze critical aspects.

Source code:

main.c:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    FILE *infile;
    int gsize, tmp;
    infile = fopen("testfile.txt", "r");
    if (infile == NULL)
    {
        printf("Failed.\n");
        return 1;
    }
    while ((fscanf(infile, "%d", &gsize)) != EOF)
    {
        int sum = 0;
        int cnt = 0;
        int ave;
        for (int i = 1; i <= gsize; i++)
        {
            fscanf(infile, "%d", &tmp);
            if (tmp % 3 != 0)
            {
                sum += tmp;
                cnt++;
            }
        }
        if (cnt != 0){
            ave = sum / cnt;
```

```
            printf("The average of this group is %d.\n", ave);
        }else{
            printf("This group has no corresponding number.\n");
        }
    }
    fclose(infile);
    return 0;
}
```

testfile.txt:

4 10 20 30 40

3 50 60 70

5 80 90 100 110 120

Program output:

```
The average of this group is 23.
The average of this group is 60.
The average of this group is 96.
```

Analysis:

1. **Key Challenges & Solutions**

 - Learn about the function *fscanf()* to get the data in another file.

 - Know that the pointer of *fscanf()* is sequent so it can be used as the terminal condition of the outer loop.

2. **Result Validation**

 - When reading the file, we need to check whether successfully read. If not we need return error information.

 - When calculating the average, we need to check whether the denominator is zero.

 - Note that break the connection between the files at last.

Conclusion:

When reading the file, we need to check whether the operation is successful and make a error judgement. When using the nested loop to input the file, the EOF can be used as the terminal condition of outer loop and the function *fscanf()* can be used to sequentially input the data in another file.